2×2 random matrices: "spacing distribution"

$$X = \begin{pmatrix} x_1 & x_3 \\ x_3 & x_2 \end{pmatrix} \qquad \begin{array}{l} x_1, x_2 \overset{i.i.d}{\sim} N(0,1) \\ x_3 \sim N(0, \frac{1}{2}) \end{array}$$

p.d.f of the spacing $S = \lambda_2 - \lambda_1$

$$\lambda^2 - (\mathrm{Tr}\, X)\lambda + \det X = 0$$

$$\lambda^2 - (x_1 + x_2)\lambda + (x_1 x_2 - x_3^2) = 0$$

$$\lambda_{1,2} = \frac{1}{2}\left[x_1 + x_2 \pm \sqrt{(x_1+x_2)^2 - 4(x_1x_2 - x_3^2)} \right]$$

$$S = \lambda_2 - \lambda_1 = \sqrt{(x_1+x_2)^2 - 4(x_1x_2 - x_3^2)}$$

$$P(s) = \int_{-\infty}^{\infty} \frac{dx_1}{\sqrt{2\pi}} \frac{dx_2}{\sqrt{2\pi}} \frac{dx_3}{\sqrt{2\pi}} e^{-\frac{1}{2}(x_1^2 + x_2^2)} e^{-x_3^2} \delta\left(s - \sqrt{(x_1+x_2)^2 - 4(x_1x_2 - x_3^2)}\right)$$

$$\begin{cases} x_1 - x_2 = r\cos\theta \\ 2x_3 = r\sin\theta \\ x_1 + x_2 = \psi \end{cases} \Longrightarrow \begin{cases} x_1 = \dfrac{r\cos\theta + \psi}{2} \\ x_2 = \dfrac{\psi - r\cos\theta}{2} \\ x_3 = \dfrac{r}{2}\sin\theta \end{cases}$$

$$|J| = \left| \begin{pmatrix} \frac{\partial x_1}{\partial r} & \frac{\partial x_1}{\partial \theta} & \frac{\partial x_1}{\partial \psi} \\ \frac{\partial x_2}{\partial r} & \frac{\partial x_2}{\partial \theta} & \frac{\partial x_2}{\partial \psi} \\ \frac{\partial x_3}{\partial r} & \frac{\partial x_3}{\partial \theta} & \frac{\partial x_3}{\partial \psi} \end{pmatrix} \right| = \left| \begin{pmatrix} \frac{\cos\theta}{2} & -\frac{r}{2}\sin\theta & \frac{1}{2} \\ -\frac{\cos\theta}{2} & \frac{r}{2}\sin\theta & \frac{1}{2} \\ \frac{\sin\theta}{2} & \frac{r}{2}\cos\theta & 0 \end{pmatrix} \right| = -\frac{r}{4}$$

$$= \frac{1}{2\pi\sqrt{2\pi}} \frac{1}{4} \int_0^{\infty} dr \int_0^{2\pi} d\theta \int_{-\infty}^{\infty} d\psi \; r\,\delta(s - r) \times$$

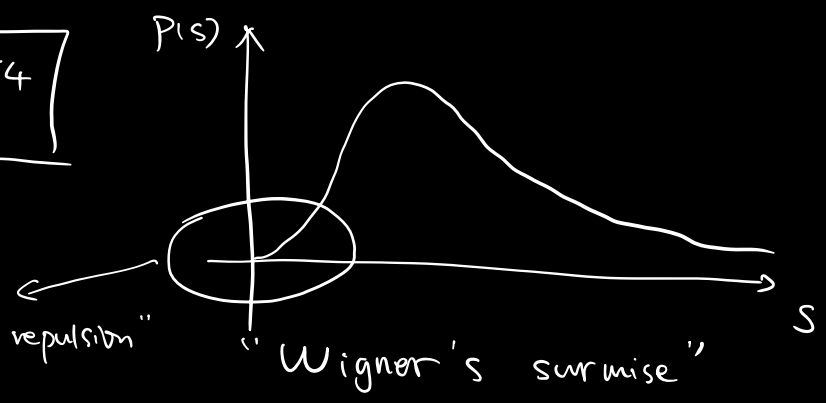
$$\exp\left\{ -\frac{1}{2}\left[\left(\frac{r\cos\theta + \psi}{2}\right)^2 + \left(\frac{\psi - r\cos\theta}{2}\right)^2 + 2\frac{r^2}{4}\sin^2\theta \right] \right\}$$

$$= \frac{1}{2\pi\sqrt{2\pi}} \frac{1}{4} s\,\theta(s) \int_0^{2\pi} d\theta \int_{-\infty}^{\infty} d\psi$$

$$\exp\left[-\frac{1}{2}\left(\frac{s^2\cos^2\theta + \psi^2 + 2s\psi\cos\theta}{4} + \frac{\psi^2 + r^2\cos^2\theta - 2s\psi\cos\theta}{4} + \frac{s^2}{2}\sin^2\theta\right)\right]$$

$$= \frac{1}{2\pi\sqrt{2\pi}} \frac{1}{4} s\,\theta(s)\, e^{-\frac{s^2}{4}} \int_0^{2\pi} d\theta \int_0^{+\infty} d\psi \exp\left(-\frac{\psi^2}{4}\right)$$

$$\Rightarrow \boxed{P(s) = \theta(s)\,\frac{s}{2}\, e^{-s^2/4}}$$



* even if the entries are independent random variables, the eigenvalues are <u>not</u>.

The j.p.d.f of eigenvalues of an $N \times N$ Gaussian matrix is

$$P(x_1, \dots, x_N) = \frac{1}{Z_{N,\beta}} e^{-\frac{1}{2}\sum_{i=1}^{N} x_i^2} \prod_{j<k} |x_j - x_k|^{\beta}$$

where $Z_{N,\beta} = (2\pi)^{\frac{N}{2}} \prod_{j=1}^{N} \frac{\Gamma(1 + j\beta/2)}{\Gamma(1+\beta/2)}$, $\beta = 1, 2, 4$ Dyson index

GOE : $\beta = 1$

GUE : $\beta = 2$

GSE : $\beta = 4$

kills any configuration of eigenvalues $\{x\}$ where some x_j's are "big"

kills configurations where two eigenvalues get "too close" to each other.

这一项使这些 eigenvalues 强烈地 non-independent !

* $\beta = 1$, $N = 2$ 就回到了一开始 2×2 对称矩阵的情形

Q: From j.p.d.f of eigenvalues $p(x_1, \dots, x_N)$, how do I compute the shape of the histograms of the $N \times T$ eigenvalues, for large T?

A: All you have to do is to take the marginal:

$$p(x) = \int \cdots \int dx_2 \cdots dx_N \, p(x, x_2, \dots, x_N)$$

$$n(x) = \frac{1}{N} \sum_{i=1}^{N} \delta(x - x_i)$$

$$\langle n(x) \rangle = \int \cdots \int dx \, p(x_1, \dots, x_N) \, n(x) = \frac{1}{N} \sum_{i=1}^{N} \int \cdots \int dx \, p(x_1, \dots, x_N) \, \delta(x - x_i)$$

$= p(x)$ called <u>spectral density</u>

Q: For large N, what does $p(x)$ for Gaussian ensemble look like?

A: $$\lim_{N \to \infty} \sqrt{\beta N} \, p(\sqrt{\beta N} \, x) = P_{sc}(x)$$

where $P_{sc}(x) = \frac{1}{\pi} \sqrt{2 - x^2}$ <u>Wigner's semicircle law</u>

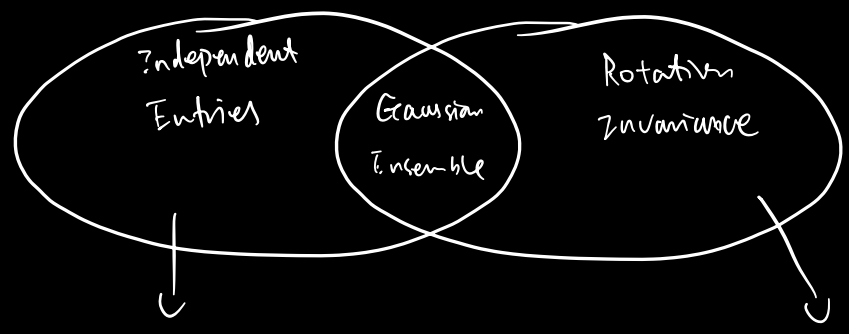
Layman's classification:

1. Independent entries
2. Rotational invariance

$$H' = U H U^{-1}$$

$$p(H) \, dH_{11} \cdots dH_{NN} = p(H') \, dH'_{11} \cdots dH'_{NN}$$

$$\Rightarrow \begin{cases} p(H) = p(H') \\ dH = dH' \end{cases} \Rightarrow p(H) = \varphi(\mathrm{Tr} H, \mathrm{Tr} H^2, \dots, \mathrm{Tr} H^N)$$



$$P(H) \propto \prod_{i=1}^{N} f_i(H_{ii}) \prod_{i<j} f_{ij}(H_{ij}) \qquad P(H) = P(UHU^{-1})$$

The Fluid semicircle

Take the j.p.d.f for the Gaussian Ensemble

$$p(x_1, \dots, x_N) = \frac{1}{Z_{N,\beta}} e^{-\frac{1}{2}\sum_{i=1}^{N} x_i^2} \prod_{j<k} |x_j - x_k|^\beta$$

rescale the eigenvalues : $x_i \longrightarrow x_i \sqrt{\beta N}$

$$\Rightarrow Z_{N,\beta} = C_{N,\beta} \int_{\mathbb{R}^N} \prod_{j=1}^{N} dx_j \, e^{-\frac{\beta}{2} N \sum_{i=1}^{N} x_i^2} \prod_{j<k} |x_j - x_k|^\beta$$

$$= C_{N,\beta} \int_{\mathbb{R}^N} \prod_{j=1}^{N} dx_j \, e^{-\beta N^2 V[x]}$$

where $C_{N,\beta} = (\sqrt{\beta N})^{N + \beta N(N-1)/2}$

$$V[x] = \frac{1}{2N} \sum_i x_i^2 - \frac{1}{2N^2} \sum_{i \neq j} \ln|x_i - x_j|$$

energy term